Santiago Santos

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Experience

Game Programmer | Internship

May 2026 – August 2026

- Developed core Unreal Engine 5 systems for a nursing simulation, owning **inventory**, **save/load**, **pickup**, and **interaction** from architecture through a shipped build using Blueprints and C++.
- Ran **manual** and **visual testing** for UI/UX systems flow end-to-end throught the façade interface, checking widgets and settings update correctly, while information is readable and consistent.
- Built a multi-purpose interaction system in **Blueprints** and **C++** that replaces per-frame line traces across gameplay systems with a single shared trace fired every **0.1 seconds**, reducing interaction queries by **80%.**
- Architected it event-driven using **component pattern** and **observer pattern**, **eliminating 10+ hard dependencies** between systems, which allows adding new elements without modifying existing code.
- Performed **integration testing** on the item pickup system to verify behaviour at inventory system capacity limits, **reproducing** and **fixing** bugs before shipping.

Technical Skills

- **Languages:** C++, C#, JavaScript, Blueprints (Unreal Engine 5), SQL, Java, HTML/CSS.
- **Engines & Graphics:** Unreal Engine 5 (Blueprints & C++), Unity, Three.js, Fabric API.
- **Tools:** Perforce, Jira, Rider, VS Code, Visual Studio, Git, GitHub, Maya, Trello, Microsoft Teams, Google Workspace.
- **Practices:** Time Management, Team Collaboration, Adaptability, Ownership, Clear Communication, S.O.L.I.D. Principles.
- **Languages (spoken):** French (Advanced), English (Fluent), Spanish (Fluent).

Projects

Sentiment – Best Audio & Participant's Pick Behaviour Interactive Game Jam 2026 Montreal, QC

Behaviour Interactive *April 2026*

- Programmed the **core gameplay loop** and all **state-driven** actions using **Blueprints** and **C++** in Unreal Engine 5.
- Implemented the full game's **audio system** end to end, from cue setup through Blueprint integration.
- Managed the project's full **QA** pass, **reproducing**, **triaging** and **fixing** gameplay bugs.

The Cycle – Best Gameplay Design Winner McGameJam 2026 Montreal, QC

McGill University *January 2026*

- Programmed and tested **state-based AI systems** driving enemy behavior and **gameplay challenges** in UE5.
- Owned and integrated **all game audio** using **Sound Cues** to enhance gameplay immersion.
- Wired together Sound Cues to **all state transitions** including **player detection** and **spatial audio**.

Killer Journalist – 1st Place & Best Art MGD Jam 2025 Montreal, QC

Indie Asylum *October 2025*

- Developed **over 80%** of the **UI systems** using **Blueprints** in **Unreal Engine 5.**
- Acted as a key **integrator**, connecting **gameplay**, sound, **UI**, and **art** to ensure project cohesion and **high quality.**
- Reviewed and **tested** the prototype, pointing bugs, and **testing** edge-cases scenarios.

Lavender Biome – A Minecraft Mod Montreal, QC

LaSalle College – School Project *September 2025 – December 2025*

- Developed a **Java-based Minecraft mod using the Fabric API**, extending the existing Minecraft codebase to introduce **10+ custom gameplay systems**, including new blocks, items, biome elements, and interactive world mechanics.
- Implemented **procedural world generation systems** for a custom biome, designing algorithms to spawn new ores, vegetation, and environmental assets while integrating seamlessly with Minecraft's existing terrain generation pipeline.

Education

DCS – Computer Science | Video Game Programming Montreal, QC

LaSalle College *August 2023 – August 2026*

- **Key Courses:** Game Engine I, II & III, Graphical Environment, Data Structures, Object Oriented Programming (OOP).
- **Extracurriculars:** Public Speaking, Team Sports, bE Interactive 3rd Place Winner, MGD 1st Place Winner.